

Worksheet 1 — The Math of the Atom

High School · Grades 9–12 · Physics & Algebra II / Pre-Calc · ~45 min

Use the constants and formulas in the reference box to solve each problem. Show your work. These problems mirror the calculations real nuclear engineers, health physicists, and reactor operators perform every day.

Name: _____

Date: _____

Class: _____

Constant / Formula	Value / Expression
Speed of light, c	3.00×10^8 m/s
1 atomic mass unit (u)	1.6605×10^{-27} kg
1 MeV	1.602×10^{-13} J
Avogadro's number	6.022×10^{23} /mol
Mass–energy equivalence	$E = mc^2$
Radioactive decay	$N(t) = N_0 \cdot (1/2)^{(t/T^{1/2})}$
Decay constant	$\lambda = \ln(2) / T^{1/2}$
Reactor multiplication	$k_{\text{eff}} = \text{neutrons produced} / (\text{neutrons absorbed} + \text{leaked})$

1. Mass–Energy Equivalence

A single fission of U-235 converts about 0.215 u of mass into energy. Calculate the energy released (a) in joules and (b) in MeV. Show all unit conversions.

2. Scaling Up

A 1,000-MW(e) reactor operates at ~33% thermal efficiency, so it generates about 3,000 MW of thermal power. Using your answer from Problem 1, estimate the number of U-235 fissions per second required to sustain that thermal output.

3. Fuel Burnup

If each fission consumes one U-235 atom (mass ≈ 235 u), how many kilograms of U-235 are fissioned per day in the reactor from Problem 2? Compare this to the $\sim 9,000$ metric tons of coal a comparable coal plant burns per day.

4. Half-Life Decay

Iodine-131 (a fission product used in medical imaging) has a half-life of 8.02 days. If a hospital receives a 200 mCi (millicurie) source, how much activity remains after 30 days? Express your answer in mCi.

5. Decay Constant

Compute the decay constant λ for Cesium-137 ($T_{1/2} = 30.17$ years) in units of per second. Why is this isotope a major concern in long-term spent-fuel storage?

6. Reactor Criticality

A reactor has $k_{\text{eff}} = 1.002$. (a) Is the reactor subcritical, critical, or supercritical? (b) If each neutron generation lasts ~ 0.1 s (delayed-neutron-dominated), by what factor does the neutron population grow in

60 seconds?

7. Energy Density Comparison

Using your Problem 3 result, calculate the specific energy of U-235 fission in MJ per kilogram and compare to gasoline (~46 MJ/kg). Express the ratio.

8. Dose Estimation (Real-World Application)

Background radiation in the US averages ~3.1 mSv/year. A single chest CT scan is ~7 mSv. A cross-country flight is ~0.04 mSv. Living next to a nuclear plant adds <0.01 mSv/year. Rank these from largest to smallest and discuss which dose source most people underestimate.

Stretch Challenge — College Preview

The four-factor formula approximates k_{∞} in a thermal reactor: $k_{\infty} = \eta \cdot f \cdot p \cdot \epsilon$, where η is the reproduction factor, f the thermal utilization, p the resonance escape probability, and ϵ the fast fission factor. Research each term and write a one-paragraph explanation of how a reactor designer would increase k_{∞} while preserving negative temperature feedback.